

Adversarial Constraint Calculus

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1 Introduction

2 Formal Definition

A proof consists of an object p such that

$$p = (\Gamma, \Sigma)$$

where Γ and Σ are sets of persistent proof context and mutable state respectively such that

$$\Gamma \subseteq \Gamma'$$

$$\Sigma \rightarrow \Sigma'$$

We can assert that a proof p shows ϕ as $p \vdash \phi$ or more formally $\Gamma \vdash \phi$ for some $p = (\Gamma, \Sigma)$. Similarly we can assert that a proof p models ϕ as $p \models \phi$ iff more formally $\Sigma \vdash \phi$ for some $p = (\Gamma, \Sigma)$. A property ϕ is sound under a proof p iff $p \vdash \phi \wedge p \models \phi$.

Instructions are total or partial functions $f : (\Gamma, \Sigma) \rightarrow (\Gamma', \Sigma')$ subject to $\Gamma' = \Gamma \cup \Delta_f(\Gamma, \Sigma)$. These act as parametric theorems which may construct a proof object from another proof object, or may return another parametric theorem.

Failure of any instruction or assertion in a proof fails the entire proof. We represent failure of a proof or assertion as $\perp$ such that

$$f(p) = \begin{cases} (\Gamma', \Sigma') & \text{if preconditions are satisfied} \\ \perp & \text{otherwise} \end{cases}$$

A construction rule constitutes addition of information to the context of a proof.

$$\text{construct}_\phi(\Gamma, \Sigma) = (\Gamma \cup \phi, \Sigma)$$

We may similarly apply a state transition to a proof state.

$$\text{update}_u(\Gamma, \Sigma) = (\Gamma, u(\Sigma))$$

Where $u : \Sigma \rightarrow \Sigma$. We notate conditional construction of b if a as an implication $a \rightarrow b$ and conditional update on b if a as an implication $a \Rightarrow b$. Naturally $(f \circ g)(p) = f(g(p))$.

3 Operational Semantics

An instruction f constitutes theorem application on p , and undergoes a transition relation

$$(\Gamma, \Sigma) \xrightarrow{f} (\Gamma', \Sigma') \text{ iff } f(\Gamma, \Sigma) = (\Gamma', \Sigma')$$

Execution is traced for $p_0 \xrightarrow{f_1} p_1 \xrightarrow{f_2} \dots \xrightarrow{f_n} p_n$ such that

$$\Gamma_0 \subseteq \Gamma_1 \subseteq \dots \subseteq \Gamma_n$$

An invariant is a property over a state $I(\Sigma)$, and is preserved if $\forall f, I(\Sigma) \xrightarrow{f} I(\Sigma')$. An invariant is certified if $\Gamma \vdash I$, and is sound only when it is both shown and modeled.

4 Adversarial Modeling